Question 1 - Aiyagari Model

The full code for this question is attached as a zip file to the problem set, with the following files:

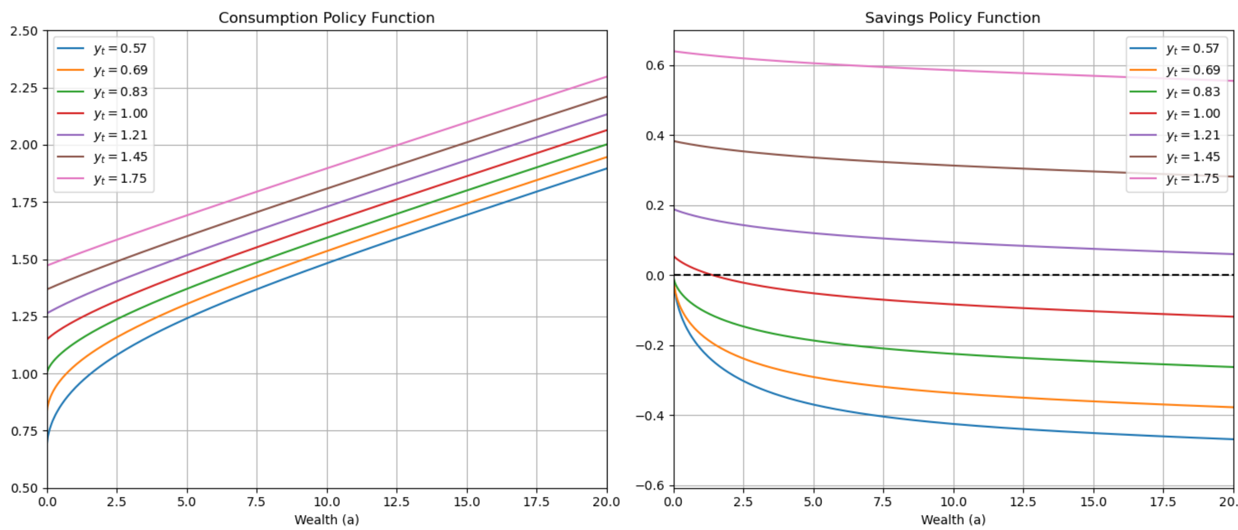
- `ha_utils.py`: File with a series of auxiliary functions used in the `aiyagari.py` file to build grids, discretize income processes, solving bellman equations, computing stationary distributions and simulating households.

- `aiyagari.py`: File with all the classes that compose the building blocks of the Aiyagari Model. It has two classes, `Households` and `Firms`, where I compute in the latter the policy functions, joint stationary distribution of income and wealth, and aggregate household variables (capital, consumption, savings, etc). The function `steady_state` in the `Households` class computes the steady state given (in partial equilibrium) given a level of wages and interest rate.

  To compute the output and the wage as functions of interest rate, I use the function within the `Firms` class. Lastly, it also includes the function `solve_steady_state`, which I use a built-in solver to find the equilibrium interest rate using the asset market equilibrium condition, $K^d(r) = K^s(r)$.

- `Main_file.ipynb`: This Jupyter Notebook has the solution to all the questions. I draw from the `ha_utils.py` and `aiyagari.py` files to compute the objects of interest of this question

Figure 1: Policy Functions



Note: Policy Functions for Consumption and Savings under the equilibrium interest rate, $r^* = 3.7\%$, and wage $w = 1.20$

Question 2 - Huggett Economy with CRRA Preferences

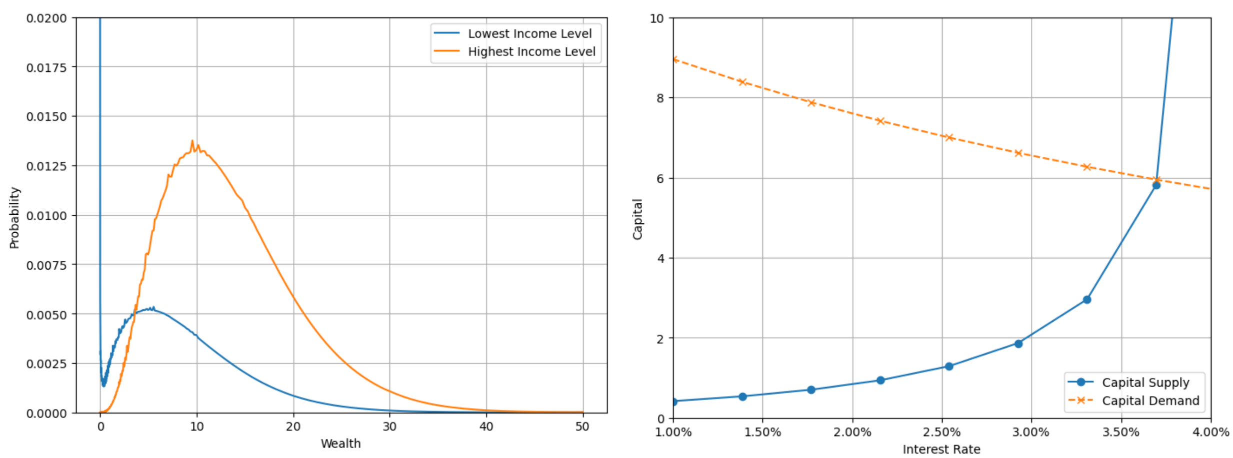
Time is discrete and indexed by $t \in \{0, 1, \ldots, \infty\}$. The economy is populated by a continuum of measure one of infinitely–lived households indexed by $i \in [0, 1]$. Preferences for household i are given by

$$U(c_0^i, c_1^i, \ldots) \;=\; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \, u(c_t^i),$$

Figure 2: Stationary Distribution & Supply and Demand of Capital

(a) Stationary Wealth Distribution Conditional On Income

(b) Supply and Demand of Capital



Note: The stationary distribution on the Panel (a) is computed with the equilibrium interest rate and wages, $r^* = 3.7\%$, $w = 1.20$

where $\beta \in (0, 1)$ is the discount factor, $c_t \geq 0$ denotes consumption at time t, $u' > 0$ and $u'' < 0$. Each household i in period t receives a stochastic idiosyncratic endowment of the consumption good $\varepsilon_t^i \in \{\varepsilon_h, \varepsilon_\ell\}$ that follows a Markov chain with transition probabilities:

$$\begin{aligned}
\pi_{hh}^{\varepsilon} &= \mathbb{P}\big\{\varepsilon_t^i = \varepsilon_h \;\big|\; \varepsilon_{t-1}^i = \varepsilon_h\big\},\\
\pi_{\ell\ell}^{\varepsilon} &= \mathbb{P}\big\{\varepsilon_t^i = \varepsilon_\ell \;\big|\; \varepsilon_{t-1}^i = \varepsilon_\ell\big\},\\
\pi_{h\ell}^{\varepsilon} &= 1 - \pi_{hh}^{\varepsilon},\\
\pi_{\ell h}^{\varepsilon} &= 1 - \pi_{\ell\ell}^{\varepsilon}.
\end{aligned}$$

Households can trade a risk–free asset a in zero–net supply with gross return R_t, and they may accumulate debt up to the exogenous borrowing limit $-\bar{b}$.

2.1 - Define the Recursive Competitive Equilibrium

Mathematical Background: We build a probability space adapted to the uncertainty generated by the idiosyncratic income shocks, and define the conditions over which the joint probability measure over the space of assets and income will have an invariant measure, λ^*. Let $\mathcal{A} = [-\bar{b}, \bar{a}]$ and $\mathcal{E} = \{\varepsilon_h, \varepsilon_\ell\}$. We have that $\mathcal{S} = \mathcal{A} \times \mathcal{E}$ defines the state space of households. To define the measure space, we take the Borel set of $\mathcal{A}$, $B_A = \mathcal{B}(\mathcal{A})$, such that we define the σ-algebra over $\mathcal{S}$, $\Sigma_S = \sigma(P(E) \times B_A)$. Therefore, $(\mathcal{S}, \Sigma_S)$ is our measure space.

In this Markov Decision Problem, as ε_t is a Markov over $(\mathcal{S}, \Sigma_S)$, by the recursive structure of the MDP, policy functions will lead the wealth decisions of households to also follow a Markov Process over $(\mathcal{S}, \Sigma_S)$. Thus, we have a joint Markov-Process for (a, ε). Recall that a Markov process will be characterized by the triple (Q, T, T^*), which are the transition function, the Markov Operator and its Adjoint Operator. The Transition operator $Q : \mathcal{S} \times \Sigma_s \to [0, 1]$ is defined as:

$$Q((a, \varepsilon), A \times E) = \mathbf{1}[a'(a, \varepsilon) \in A \times E] \sum_{\varepsilon' \in E} \pi(\varepsilon' \mid \varepsilon) \tag{1}$$

Thus, the Markov operator (assuming Q satisfies the Feller Property), is an operator $T : \mathcal{C}(\mathcal{S}) \to \mathcal{C}(\mathcal{S})$ defined as:

$$(Tf)(a,\varepsilon) = \int f(a',\varepsilon')Q((a,\varepsilon),(da',d\varepsilon')), \quad \forall\,(a,\varepsilon) \in \mathcal{S} \tag{2}$$

and the adjoint operator $T^* : \Lambda(\mathcal{S} \times \Sigma_s) \to \Lambda(\mathcal{S} \times \Sigma_s)$, where $\Lambda(\mathcal{S} \times \Sigma_s)$ is the space of probability measures over $(\mathcal{S} \times \Sigma_s)$:

$$(T^*\lambda)(A \times E) = \int_{A\times E} Q((a,\varepsilon), A \times E)\lambda(da, d\varepsilon) \tag{3}$$

We can now proceed to define the Rational Expectations Recursive Competitive Equilibrium of this Economy:

Definition ((RE) Recursive Competitive Equilibrium)**.** *Given a Measure Space $(\mathcal{S}, \Sigma_S)$ over the state-space of households, $\mathcal{S} = \mathcal{A}\times\mathcal{E}$, a Markov Chain Transition Function $\pi(\varepsilon' \mid \varepsilon)$ for the idiosyncratic Income Process, and a Triplet (Q, T, T^*), a* ***Recursive Competitive Equilibrium*** *is a set of Σ_s-measurable Value and Policy Functions $\{V^*(a,\varepsilon), a'^*(a,\varepsilon), c^*(a,\varepsilon)\}$, input prices $\{r^*\}$, aggregate quantities $\{Y^*, C^*, I^*, A^*\}$ and invariant measure $\lambda \in \Lambda(\mathcal{S}, \Sigma_S)$ such that:*

1. *Households Optimize: Given price (i) $\{r^*\}$; (ii) borrowing constraint $\underline{b}$; and (iii) a Markov-chain transition function for the idiosyncratic income process $\pi(\varepsilon' \mid \varepsilon)$, the Value and Policy Functions $\{V^*(a,\varepsilon), a'^*(a,\varepsilon), c^*(a,\varepsilon)\}$ solve the following functional equation:*

$$\begin{aligned} V^*(a,\varepsilon) = \max_{c,a'} &\left\{ u(c) + \beta \sum_{\varepsilon' \in \mathcal{E}} \pi(\varepsilon' \mid \varepsilon) u(c(a'^*(a,\varepsilon),\varepsilon')) \right\} \\ s.t.& \\ c + a' &= Ra + \varepsilon \\ a' &\geq \underline{a} \end{aligned}$$

2. *Market's Clear: Markets for Goods and Assets Clear:*
   - *Goods Market: $Y^* = C^* + I^*$*
   - *Capital Market: $0 = A^* = \int_{\mathcal{S}} a'^*(a,\varepsilon)\lambda^*(da, d\varepsilon)$*

3. *Measure Satisfies Consistency: The invariant measure $\lambda \in \Lambda(\mathcal{S}, \Sigma_S)$ is the fixed point of the Adjoint Markov Operator T^*,*

$$\lambda^*(A \times E) = (T^*\lambda^*)(A \times E) \tag{4}$$

Existence and Uniqueness: In this economy, we have two markets and one prices (By Walras' Law, we normalized the prices of goods to be 1), $\{r\}$. So in the end, checking the existence and uniqueness of the recursive stationary equilibrium of the economy boils down to checking whether the excess asset demand function $z : \mathbb{R} \to \mathbb{R}$

$$z(r) = A(r) - 0 = \int_{A\times E} a'^*(a,\varepsilon; r)\lambda^*(da, d\varepsilon; r) \tag{5}$$

has a unique solution $z(r^*) = A(r^*) = 0$. To show this, we show that the function $A(r)$ is continuous, monotonic and crosses once the 0 value:

1. **Existence of the Invariant Measure**: To ensure that the function $A(r)$ is well defined, we must ensure the existence of the invariant measure λ^*. This follows from theorem 12.10 of Stokey, Lucas and Prescott:

Theorem 1. *If $Z \subseteq \mathbb{R}^n$ is compact and P has the Feller property, then there exists a probability measure that is invariant under the transition function Q*

In our case, $Z = \mathcal{A} \times \mathcal{E}$ is compact as it is the cartesian product of a compact $\mathcal{A}$ and a countable set $\mathcal{E}$. This follows as, under the assumption of $\beta(1+r) < 1$, it was shown in class that by the supermartingale convergence theorem, the stochastic process for assets and consumption is bounded above. Therefore, we can define $\mathcal{A}$ as a compact set. The feller property of the Markov Operator $(Tf)(a, \varepsilon)$ follows from the continuity of the policy function $a'(a, \varepsilon)$ (which follows from Berge's Theorem of the Maximum) and its boundedness, as the state space $\mathcal{S}$ is bounded. Lastly, by the Theorem 9.14 in SLP, if ε is a discrete Markov Chain, $\mathcal{E}$ is countable, $\mathcal{S} = \sigma(P(\mathcal{E}) \times B_A)$ and a' is continuous, then Q satisfies the Feller Property

2. **Uniqueness of the Invariant Measure**: Recall the Theorem 12.12 of Stokey, Lucas and Prescott:

Theorem 2 (Uniqueness of the Invariant Measure under P)**.** *Let $Z = [a, b] \subset \mathbb{R}^n$. If P is (I) monotone; (II) has the Feller Property; and (III) satisfies the Monotone Mixing Condition Assumption, then P has a unique invariant probability measure λ^*, and $T^{*n}\lambda_0 \to \lambda^*$, for any given $\lambda_0 \in \Lambda(Z, \mathscr{L})$*

Condition (III) is satisfied as long as we assume that the transition probabilities $\pi_{\ell\ell}, \pi_{hh} \in (0, 1)$, ensuring there is mixing in the Markov chain (any state can be reached $\mathbb{P}$-a.s. with a finite period of time). We already checked that Q satisfies the Feller property, so the last condition is to check (I), which is that P is monotone: As (I) the markov chain is monotone; and (II) the next-period asset holdings $a'(a, \varepsilon)$ is increasing in both arguments, then the Markov operator T will map increasing functions f into (Tf) non-decreasing ones. Thus, Q is monotone

3. **Continuity with respect to** r: In the following exercise, we have that (indirectly), the policy functions depend on the parameters of the Income Fluctuation Problem. Specifically, as we are interested in $A(r)$, we make the dependence of the policy function and the invariant measure with respect to r explicit in equation (5). To see whether $A(r)$ varies continuously in r, we invoke the following Theorem 12.13 from Stokey, Lucas and Prescott:

Theorem 3 (Theorem 12.13 - SLP)**.** *Assume that*

a. $\mathcal{S}$ is compact;

b. if $\{(s_n, \theta_n)\}$ is a sequence in $\mathcal{S} \times \Theta$ converging to (s_0, θ_0), then the sequence $\{Q_{\theta_n}(s_n, \cdot)\}$ in $\Lambda(S, \mathscr{L})$ converges weakly to $Q_{\theta_0}(s_0, \cdot)$; and

c. for each $\theta \in \Theta$, T^*_θ has a unique fixed point $\lambda_\theta \in \Lambda(S, \mathscr{L})$.

If $\{\theta_n\}$ is a sequence in Θ converging to θ_0, then the sequence $\{\lambda_{\theta_n}\}$ converges weakly to λ_{θ_0}.

In the context of the following Bewley model, continuity of the policy function for next period assets $a'^*(a, \varepsilon)$ together with the previously defined conditions is enough to guarantee weak convergence of the measure λ_{r_n}. Therefore, we can now turn to the continuity of $A(r)$. Take a sequence $\{r_n\}_{n=1}^{\infty}$:

$$A(r_n) = \int_{A \times E} a'^*(a, \varepsilon; r_n)\lambda^*(da, d\varepsilon; r_n) \xrightarrow{n \to \infty} \int_{A \times E} a'^*(a, \varepsilon; r^*)\lambda^*(da, d\varepsilon; r^*) = A(r^*) \tag{6}$$

4. **Monotonicity**: There are no results on the monotonicity of the aggregate supply of capital with respect to r, so uniqueness is never guaranteed. Intuitively, a higher r has both income and substitution effects on savings: the relative dominance between the two could switch at a certain level of assets, such that the asset function $a'(a, \varepsilon; r)$ may not be monotone in r

2.2 - Autarky Economy

On this item, we consider the case in which preference are CRRA and the ad-hoc borrowing constraint is a no-borrowing constraint, $b = 0$. The implication of this assumption is that since no one can borrow and zero-net supply of bonds has to hold, nobody can save in equilibrium. Therefore, we know that in equilibrium $a'(0, \varepsilon) = 0$, and that there's no need to characterize for positive levels of assets since the stationary distribution over assets has all its mass on 0 in this special case. Moreover, we know that in equilibrium, all agents are hand-to-mouth $c(0, \varepsilon) = \varepsilon$. Define $R = 1 + r$ and $\beta = \dfrac{1}{1+\rho}$, where r is the risk-free rate of return and ρ the discount rate. The following proposition holds in this Huggett economy:

Proposition 4. *The equilibrium risk-free rate of return satisfies $r < \rho$*

Proof. Take the bellman equation in the household problem:

$$V(\varepsilon, a) = \max_{a' \geq -\phi} \{u(Ra + \varepsilon - a') + \beta \sum_{\varepsilon' \in E} V(\varepsilon', a')\pi(\varepsilon'|\varepsilon)\}$$

when solving it, we arrive at euler equation:

$$u'(c_t) \geq \beta R \mathbb{E}_t[u'(c_{t+1})]$$

Now, use the fact that in equilibrium, $c_t = \varepsilon_t$ and open the expected value on the RHS (conditional on $\varepsilon_t = \varepsilon_H$):

$$\begin{aligned}
&\Rightarrow (\varepsilon_H)^{-\gamma} \geq \beta R \left[\pi_{HH}(\varepsilon_H)^{-\gamma} + (1 - \pi_{HH})(\varepsilon_H)^{-\gamma}\right] \\
&\Leftrightarrow 1 \geq \beta R \left[\pi_{HH} + (1 - \pi_{HH}) \left(\frac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma}\right]
\end{aligned}$$

given that $\varepsilon_H > \varepsilon_L$, we have that $\left(\dfrac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma} > 1$, and thus $\left[\pi_{HH} + (1 - \pi_{HH}) \left(\dfrac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma}\right] > 1$. Therefore:

$$\begin{aligned}
&\Rightarrow 1 \geq \beta R \left[\pi_{HH} + (1 - \pi_{HH}) \left(\frac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma}\right] \\
&\Leftrightarrow \beta R \leq \frac{1}{\left[\pi_{HH} + (1 - \pi_{HH}) \left(\frac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma}\right]} < 1 \\
&\Leftrightarrow \beta R < 1 \Leftrightarrow 1 + r < 1 + \rho \Leftrightarrow r < \rho
\end{aligned}$$

□

In what follows, I assume that $1 \geq \beta R \left[\pi_{HH} + (1 - \pi_{HH}) \left(\dfrac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma}\right]$ holds at equality. Therefore, the Gross rate of the risk free asset is given by:

$$R = \frac{1}{\beta \left[\pi_{HH} + (1 - \pi_{HH}) \left(\frac{\varepsilon_H}{\varepsilon_L}\right)^{\gamma}\right]}$$

Thus, we can see that

1. R is decreasing with β, γ and $\dfrac{\varepsilon_H}{\varepsilon_L}$
2. R is increasing in π_{HH}

Question 3 - Bewley Model with Endogenous Labor Supply and Housing

Consider the following variant of the Bewley model. An individual consumer, indexed by $i \in (0,1)$, chooses consumption c_{it}, hours worked n_{it} and housing h_{it} to solve

$$\max_{c_{it},n_{it},h_{it}} \sum_{t=0}^{\infty} \beta^t u(c_{it}, n_{it}, h_{it})$$

subject to

$$P_t c_{it} + Q_t \left(h_{it+1} - (1-\delta)h_{it}\right) + B_{it+1} = W_t e_{it} n_{it} + (1 + i_{t-1})B_{it},$$

where i_{t-1} is the nominal interest rate in the economy, P_t is the price of the consumption good, Q_t is the price of housing, B_{it+1} is the consumer's holdings of financial assets, W_t is the wage rate per efficiency unit of labor, e_{it} is the consumer's endowment of labor efficiency, and n_{it} is the amount of labor supplied. The consumer's holdings of the financial asset are subject to a borrowing constraint:

$$B_{it+1} \geq -\theta Q_t h_{it+1}.$$

3.1 - Find the First Order Conditions

Define the Lagrangian of the above problem as:

$$\begin{aligned}\mathscr{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ u(c_{it}, n_{it}, h_{it}) + \lambda_t \Big[& W_t e_{it} n_{it} + (1 + i_{t-1})B_{it} \\ & - (p_t C_{it} + Q_t(h_{it+1} - (1-\delta)h_{it}) + B_{it+1}) \Big] \\ & + \gamma_t \left[-\theta q_t h_{it+1} - B_{it+1} \right] \Big\}\end{aligned}$$

The first-order conditions are given by:

$$\begin{aligned}[c_{it}] :& u_c(c_{it}, n_{it}, h_{it}) - p_t \lambda_t = 0 \\ [n_{it}] :& u_n(c_{it}, n_{it}, h_{it}) + \lambda_t W_t e_{it} = 0 \\ [B_{it+1}] :& -\lambda_t - \gamma_t + \beta \lambda_{t+1}(1 + i_t) = 0 \\ [h_{it+1}] :& -\lambda_t Q_t - \gamma_t \theta Q_t + \beta u_h(c_{it+1}, n_{it+1}, h_{it+1}) + \beta(1-\delta) q_{t+1} \lambda_{t+1} = 0\end{aligned}$$

From the first two FOCs, we can obtain a labor-supply condition:

$$\frac{u_n(c_{it}, n_{it}, h_{it})}{u_c(c_{it}, n_{it}, h_{it})} = \frac{W_t}{P_t} e_{it} \tag{7}$$

The last equation is a kind of euler equation for housing:

$$Q_t \left(\lambda_t + \gamma_t \theta \right) = \beta u_h(c_{it+1}, n_{it+1}, h_{it+1}) + \beta(1-\delta) q_{t+1} \lambda_{t+1} \tag{8}$$

whereas we obtain a similar euler equation for bond holdings by combining equations 1 and 2:

$$\frac{u_c(c_{it}, n_{it}, h_{it})}{P_t} + \gamma_t = \beta(1 + i_t) \frac{u_c(c_{it+1}, n_{it+1}, h_{it+1})}{P_{t+1}} \tag{9}$$

where $\gamma_t \geq 0$ is a Langrange multiplier that, when binding, yields the usual inequality for the constrained household as in the Income Fluctuation Problem:

$$u_c(c_{it}, n_{it}, h_{it}) > \beta(1 + r_t) u_c(c_{it+1}, n_{it+1}, h_{it+1})$$

where $1 + r_t = (1 + i_t) \frac{P_t}{P_{t+1}}$

3.2 - Rewrite the Budget Constraint

Define $A_{it} \equiv B_{it} + q_t h_{it}$ as the consumer's net worth. Then, it follows that:

$$p_t C_{it} + Q_t(h_{it+1} - (1-\delta)h_{it}) + B_{it+1} = W_t e_{it} n_{it} + (1 + i_{t-1})B_{it}$$

$$\Rightarrow \quad p_t C_{it} + \underbrace{(B_{it+1} + Q_t h_{it+1})}_{\equiv A_{it+1}} - q_t(1-\delta)h_{it} = W_t e_t n_{it} + (1 + i_{t-1})B_{it}$$

$$\Rightarrow \quad p_t C_{it} + A_{it+1} - Q_t(1-\delta)h_{it} = W_t e_t n_{it} + (1 + i_{t-1})(A_{it} - q_t h_{it})$$

$$\Rightarrow \quad p_t C_{it} + A_{it+1} = W_t e_t n_{it} + (1 + i_{t-1})A_{it} + [Q_t(1-\delta) - Q_{t+1}(1 + i_{t-1})]\, h_{it}$$

Whereas, for the borrowing constraint:

$$B_{it+1} \geq -\theta Q_t h_{it+1}$$

$$\iff \quad B_{it+1} + Q_t h_{it+1} \geq -\theta Q_t h_{it+1} + q_t h_{it+1}$$

$$\iff \quad A_{it+1} \geq (1-\theta) Q_t h_{it+1}$$

3.3 - Rescale the Budget Constraint

Rescaling the budget constraint and defining $q_t \equiv \frac{q_t}{p_t}$:

$$c_{it} + \frac{1}{P_t} A_{it+1} = \frac{W_t e_t n_{it}}{P_t} + (1 + i_{t-1}) \frac{A_{it}}{P_t} \frac{P_{t-1}}{P_{t-1}} + \left[\frac{Q_t(1-\delta) - Q_{t+1}(1 + i_{t-1})}{p_t} \right] h_{it}$$

$$\implies c_{it} + a_{it+1} = w_t e_t n_{it} + (1 + r_t) a_{it} + (q_t(1-\delta) - q_{t+1}(1 + r_{t-1}))\, h_{it}$$

as for the borrowing constraint:

$$a_{it+1} \geq (1-\theta) q_t h_{it+1}$$

Question 4 - Two-asset Huggett Economy with Endogenous Labor Supply

In this economy, there are two assets: a risk-free bond b and a risky asset, capital k. There is no market for capital, but there's a market for bonds in zero net supply. As before, the economy is populated by a continuum of measure one of infinitely-lived households indexed by $i \in [0, 1]$, that supplies optimal hours worked h_t^i and invests in bonds b_t^i whose return is $R_t = 1 + r_t$, but subject to a natural borrowing constraint $-\phi$. Moreover, each household has a firm that produces a final good y with identical technology $F(z_t^i, k_t^i, n_{it})$. This firm (or, equivalently, each household) is subject to a productivity shock z_t^i that follows a markov chain with transition probabilities $\pi(z'|z)$ and $z_{min} = 0$. The firm uses the household capital, whose return is stochastic $\tilde{r}_t = z_t^i \dfrac{\partial F}{\partial k}(z_t^i, k_t^i, n_{it}) - \delta$, and hires labor n_{it} on the labor market. Notice that the the shock z_t^i is realized in the beginning of period t, after capital k_t^i is installed but before employment n_{it} is chosen.

$$\begin{aligned}
\max_{\{c_t^i, k_{t+1}^i, b_{t+1}^i, h_t^i, n_{it}\}_{t=0}^{\infty}} \quad & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i, 1-h_t^i) \\
\text{s.t.} \quad & c_t^i + k_{t+1}^i + b_{t+1}^i = \pi_t^i + R_t b_t^i + w_t h_t^i + (1-\delta)k_t^i \\
\text{and} \quad & \pi_t^i = F(z_t^i, k_t^i, n_{it}) - w_t n_{it} \\
\text{and} \quad & b_{t+1}^i \geq -\phi \\
\text{and} \quad & k_{t+1}^i \geq 0
\end{aligned}$$

4.1 - Natural Borrowing Constraint

Definition (Endogenous/Natural Debt Limit)**.** *The Natural Debt Limit is a kind of ad-hoc borrowing constraint on the value of debt to be repayed in case which the agent receives always the worst realization of income y_{min} and sets future consumption to 0.*

$$b_{t+1}^i \geq -\frac{1}{R_t} \sum_{j=0}^{\infty} \frac{y_{min}}{R_t^{t+\tau}} \tag{10}$$

where $R_t^{t+\tau+1}(s^{t+\tau}) \equiv \prod_{s=0}^{\tau-1} R_{t+s}(s^{t+s})$ is the yield curve

In the case of this model, the natural borrowing constraint with stochastic income is defined by the equivalent of the present value of the stream of worst possible incomes for the household, that is, when $z_{t+j} = 0, \forall j \geq 0$, and that it does not consume anything $c_{t+j} = 0 \quad \forall j \geq 0$. Moreover, under the z_{min}, the household would not invest in any future capital, so we have that $k_{it+1} = 0$, and the household chooses $n_{it}^* = 0$ not to have any losses. Imposing that $F(0, k_{it}, n_{it}) = 0$, we have that:

$$\begin{aligned}
\phi =& \frac{1}{R_t} \sum_{j=0}^{\infty} \frac{w_{t+j} h_{t+j}^i}{R_t^{t+j}} \\
\Leftrightarrow b_{t+1}^i \geq& -\frac{1}{R_t} \sum_{j=0}^{\infty} \frac{w_{t+j} h_{t+j}^i}{R_t^{t+j}}
\end{aligned}$$

Assuming that the household supplies labor inelastically at the natural borrowing limit, and for simplicity that the wage and the interest rate is constant, the above inequality becomes:

$$b_{t+1}^i \geq -\frac{w}{r} \tag{11}$$

4.2 - Write The problem in Recursive Form

b. For each household, the state variables of their dynamic optimization problem are $\{z_t, b_t, k_t\}$ and the control variables are $\{h_t, n_t, c_t, k_{t+1}, b_{t+1}\}$. We can write the problem of the household in the recursive formulation as the following Bellman Equation:

$$\begin{aligned}
V_t(z, b, k) = \max_{\left\{\substack{h \geq 0,\, b' \geq -\phi, \\ n,\, k',\, b'}\right\}} & \Bigg\{ u\left(F(z, k, n) + w(h - n) + Rb + (1-\delta)k - k' - b',\, 1 - h\right) \\
& + \beta \sum_{z' \in \mathcal{Z}} P(z'|z) V_{t+1}(z', b', k') \Bigg\}
\end{aligned}$$

4.3 - Recursive Competitive Equilibrium

Mathematical Preliminaries: We proceed as in question 2.1. Let $\mathcal{S} = \mathcal{Z} \times \mathcal{B} \times \mathcal{K}$ be the state-space of households, where $\mathcal{Z}$ is a countable set, and $\mathcal{B}$ and $\mathcal{K}$ are both bounded below and above, as in the case of bonds we assume that $\beta R < 1$, so that the asymptotic path of bonds is bounded above; and in the case of capital because of decreasing marginal returns to capital (an infinite accumulation of capital would lead the return on capital to go negative, which is not optimal). Therefore, $\mathcal{B} = [-\Phi, \bar{b}]$ and $\mathcal{K} = [0, \bar{k}]$. Thus, $\mathcal{S}$ is compact. Define $\Sigma_S = \sigma(P(\mathcal{Z}) \times B_K \times B_B)$ as the σ-algebra.

The Triplet which dictates the joint Markov-Process for capital, bonds and income is given by (Q, T, T^*). The transition function $Q : \mathcal{S} \times \Sigma_s \to [0, 1]$ is:

$$Q((z, b, k), Z \times B \times K) = \mathbf{1}[b'(z, b, k) \in Z \times B \times K]\mathbf{1}[k'(z, b, k) \in Z \times B \times K] \sum_{z' \in \mathcal{Z}} \pi(z' \mid z) \tag{12}$$

Thus, the Markov operator (assuming Q satisfies the Feller Property), is an operator $T : \mathcal{C}(\mathcal{S}) \to \mathcal{C}(\mathcal{S})$ defined as:

$$(Tf)(z, b, k) = \int f((z', b', k')Q((z, b, k), dz', dk', db'), \quad \forall\, (z, b, k) \in \mathcal{S} \tag{13}$$

and the adjoint operator $T^* : \Lambda(\mathcal{S} \times \Sigma_s) \to \Lambda(\mathcal{S} \times \Sigma_s)$, where $\Lambda(\mathcal{S} \times \Sigma_s)$ is the space of probability measures over $(\mathcal{S} \times \Sigma_s)$:

$$(T^*\lambda)(Z \times B \times K) = \int_{A \times E} Q((z, b, k), Z \times B \times K)\lambda^*(dz, dk, db) \tag{14}$$

We can now proceed to define the Rational Expectations Recursive Competitive Equilibrium of this Economy:

Definition (Recursive Competitive Equilibrium)**.** *Given a Measure Space $(\mathcal{S}, \Sigma_S)$ over the state-space of households, $\mathcal{S} = \mathcal{Z} \times \mathcal{B} \times \mathcal{K}$, a Markov Chain Transition Function $\pi(z' \mid z)$ for the idiosyncratic Income Process, and a Triplet (Q, T, T^*), a* ***Recursive Competitive Equilibrium*** *is (I) a set of Σ_s-measurable Value and Policy Functions $\{c(z, b, k), k'(z, b, k), b'(z, b, k), n(z, b, k), h(z, b, k); v(z, b, k)\}$, (II) Σ_s-measurable prices $\{R, w\}$; (III) a Σ_s-measurable allocation $\{Y, C, H, A, K\}$ of output,aggregate consumption, aggregate employment, aggregate risk-free assets and aggregate capital; and (IV) an invariant measure $\lambda^*(z, b, k) \in \Lambda(\mathcal{S} \times \Sigma_s)$ such that:*

1. *Given (I) prices $\{R, w\}$, (II) borrowing constraints, and (III) the markoc-chain transition function for the idiosyncratic income $\pi(z' \mid z)$, the policy and value functions $\{c(z, b, k), k'(z, b, k), b'(z, b, k), h(z, b, k); V(z, b, k)\}$ solve the household's problem (consumption, optimal portfolio and labor supply)*

2. *Given prices $\{R, w\}$, the policy function $n(z, k)$ maximizes the firm's profit*

3. *The Goods, Labor and Risk-free asset market clear:*

Goods: $Y = C + I$

$$\int_{\mathcal{B}} \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} F(z, k, n(z, k, b))\lambda^*(dz, db, dk) + \int_{\mathcal{B}} \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} k'(z, b, k)\lambda^*(dz, db, dk)$$
$$= \int_{\mathcal{B}} \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} c(z, b, k)\lambda^*(dz, db, dk) + (1 - \delta) \int_{\mathcal{B}} \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} k\lambda^*(dz, db, dk)$$

Labor: $H = \int_{\mathcal{B}} \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} h(z, b, k)\lambda^*(dz, db, dk) = \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} n(z, k)\lambda^*(dz, db, dk)$

Risk-Free assets: $A(R) = \int_{\mathcal{B}} \int_{\mathcal{K}} \sum_{z \in \mathcal{Z}} b'(z, b, k)\lambda^*(dz, db, dk) = 0$

4. *Measure Satisfies Consistency: The invariant measure $\lambda^*(z, b, k)$ is the fixed-point of the Adjoint Markov Operator T^*,*

$$\lambda^*(Z \times B \times K) = (T^*\lambda^*)(Z \times B \times K) \quad \forall\, Z \times B \times K \in \mathcal{S} \tag{15}$$

4.4 - Existence and Uniqueness of the RCE

Existence and Uniqueness of the RCE can be shown just as in the case of question 2. As in the problem 2, uniqueness cannot be guaranteed as we cannot guarantee that the supply of assets is monotonic in the interest rate R

4.5 - Creating an Additional Market

The solution to this problem depends on the interpretation that we take on when the idiosyncratic productivity shock is observed by households

- If we assume that households observe shocks and then make investment decisions k_t^i, then what happens is that all firms choose to invest on the firm(s) whose productivity is (are) $z = z_{max}$. Therefore, define Θ as the distribution of the productivity shocks, such that the state variables of this problem become $\{\Theta, b\}$. Moreover, since all household invest on the same firms, with the same technology and productivity shock, there is no heterogeneity of household and the problem collapses into a representative agent problem.
- Now assume, as we did on rest of the exercise, that first households invest on capital and then they observe the shock z_t. Then given the distribution Θ on a period t, households will invest on the firm(s) whose expected return $\mathbb{E}[r_{t+1}|z_t]$ is/are the highest. Therefore, the state variables of the problem are $\{\Theta, k, b\}$ and all households invest on the same firm(s). Thus, there will be no heterogeneity of household and and the problem collapses into a representative agent problem.